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Lab 1 Report - Group 5 - Experimental Fluid Dynamics

Wind Tunnel: Force Measurements

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Academic year: 2024-2025

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1. Introduction

This report details a wind tunnel experiment conducted to calibrate and analyze the aerodynamic properties of a specific airfoil profile. Initially, the wind tunnel was calibrated with no objects inside, allowing for baseline measurements of pressure differences across the test section. To achieve accurate results, an external force balance was calibrated with added weights, allowing for systematic adjustments in calibration at different angles (0°, 45°, and 90°). After establishing the calibration framework, an airfoil was introduced to the tunnel, and measurements were taken across various angles of attack to simulate real-world aerodynamic conditions.

The primary objective of this experiment is to compute aerodynamic coefficients, including the lift coefficient (C_L), drag coefficient (C_D), and moment coefficient (C_M), which are essential for characterizing airfoil performance. Additionally, corresponding forces such as lift, drag, and moment were derived. This data was then used to construct the polar plot of the airfoil, providing a visual representation of its aerodynamic efficiency. Key calculations included developing a calibration matrix in MATLAB and determining the airflow velocity through dynamic and static pressure data. The results serve to enhance understanding of the aerodynamic behavior of airfoils under different conditions, offering valuable insights for applications in aerodynamics and fluid mechanics.

2. Experiment Description and Setup

The experiment was conducted in a small-scale, closed-circuit wind tunnel equipped with a test section specifically designed for aerodynamic testing. The tunnel's design allowed a controlled flow of air through the test section, with adjustable airflow speeds to simulate various aerodynamic conditions. The test section was transparent on three sides, enabling clear observation and the placement of pressure taps and other sensors.

2.1. Instrumentation and Measurement Systems

The instrumentation for this experiment included sensors to measure differences of pressures, force, and moment data. Key components included:

- **Force Balance:** A three-component force balance was used to capture the lift, drag, and pitch moment forces exerted on the airfoil. The force balance was calibrated before the experiment to ensure it could accurately measure forces at different angles.

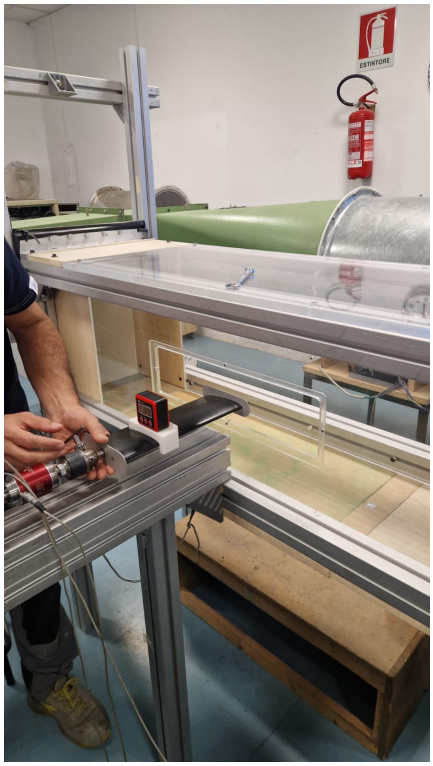

Figure 1: Attachment of the calibration system with the airfoil

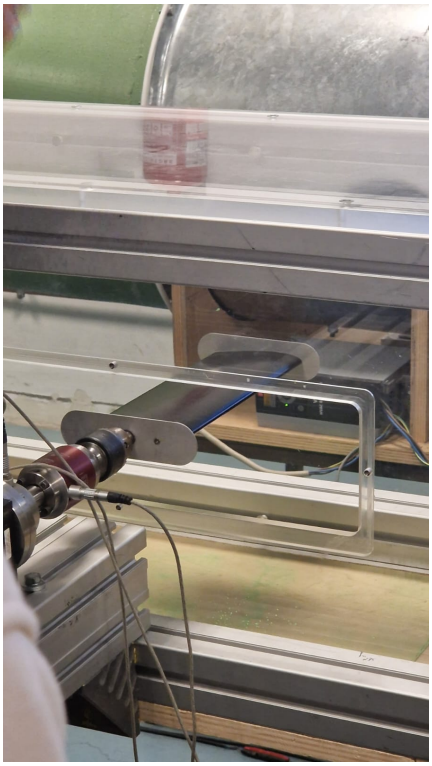

Figure 2: Installation of the airfoil inside the wind tunnel

- **Pressure Differential Sensors (Channels CH0, CH1, and CH2):** Three channels recorded specific pressure differences in the wind tunnel, which were crucial for calculating dynamic pressure and understanding the airflow characteristics:
  - **CH0:** Measures the difference between the static pressure and a pressure point $P2$, located before the airfoil at the outlet of the convergent section of the tunnel.
  - **CH1:** Measures the difference between the total pressure and $P2$, providing additional information on the flow conditions.
  - **CH2:** Measures the pressure difference between $P2$ and $P1$, where $P1$ is positioned closer to the airfoil in the narrowest section of the tunnel.

  By calculating CH1 − CH0, the dynamic pressure of the airflow was determined, which is fundamental for calculating flow velocity and other aerodynamic coefficients.

- **Angle of Attack Mechanism:** An adjustable mounting system was employed to control the airfoil's angle of attack precisely. The airfoil was positioned at various angles (between -10°and 13°) to evaluate its aerodynamic behavior across a range of flow conditions, capturing the boundary layer detachment.

2.2. Experimental Procedure

With the instrumentation and calibration complete, the experiment proceeded as follows:

1. **Tunnel Speed Adjustment:** The wind tunnel was set to a constant speed, selected based on the desired Reynolds number for the airfoil, which is critical for simulating realistic flow conditions. All measurements were conducted with $Re \approx 1.5 \cdot 10^5$.

2. **Initial Calibration Measurement with Airfoil in Place:** The airfoil was initially placed in the test section with a zero angle of attack to confirm that the force and pressure measurements aligned with calibration data. This ensured that the experimental setup was correct and that the instruments

were reading accurately.

3. **Data Collection for Varying Angles of Attack:** The angle of attack was varied in incremental steps (2 increments/decreases) to capture data across a range of flow interactions with the airfoil. At each angle, the following data were collected:
   - **F_x and F_y Forces:** Measured by the force balance, these values were used to calculate the lift coefficient (C_L) and drag coefficient (C_D) of the airfoil.
   - **Pressure Distribution:** Pressure readings from sensors along the wind tunnel provided data to calculate the dynamic pressure, and therefore, to compute the velocity of the air flow.
   - **Moment Measurement:** The pitching moment acting around the airfoil's quarter-chord was recorded to compute the moment coefficient (C_M), which describes the aerodynamic stability characteristics of the airfoil.

4. **Data Logging:** All measurements were recorded and stored for later analysis. A brief check of the data was conducted in MATLAB to ensure that each dataset was complete and accurately reflected the experimental setup.

3. Equipment Calibration

In order to produce adequate results with the setup heretofore described, it is necessary to calibrate different instruments before proceeding with the experiments. This section will describe the calibration procedure followed for the wind tunnel itself and for the force balance used to determine the lift, drag and pitching moment coefficients of the airfoil.

Regarding the wind tunnel, the goal of the calibration is to relate the pressure difference upstream and downstream of the convergent section before the test chamber, denoted as $p_1 - p_2$, with the dynamic pressure at the chamber itself. The former can be seen as a control parameter for the tunnel, while the latter must be known to non-dimensionalize the forces and moments measured throughout the experiment, but

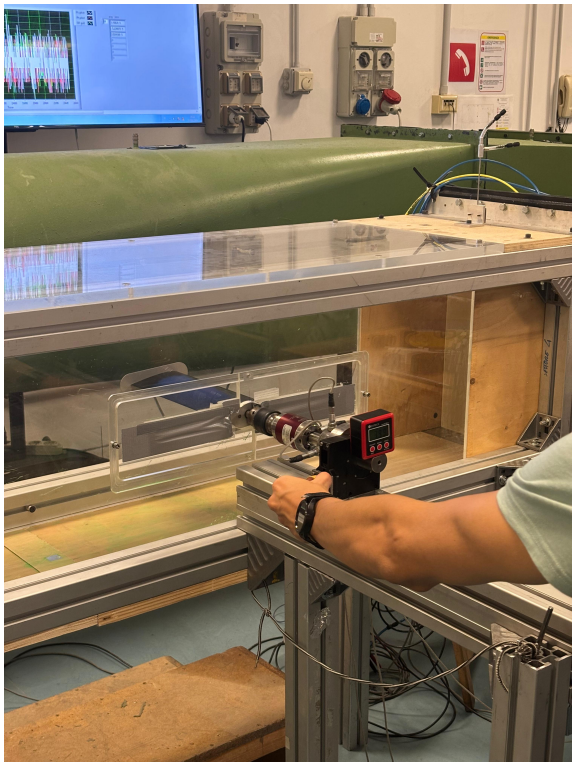

Figure 3: Positioning a new airfoil angle of attack

it is not trivial to reliably measure them while the chamber is occupied by the model. Consequently, a series of runs are performed on an empty wind tunnel, measuring three pressure differences: the aforementioned $p_1 - p_2$, $p_s - p_2$ and $p_t - p_2$, where p_s and p_t are the static and total pressure in the test chamber, respectively. Note the measurement system gives the outputs in Volts, which can be converted to pressure by means of the device's sensitivity of 200 Pa/V.

It is common practice to relate dynamic pressure and $p_1 - p_2$ linearly when calibrating wind tunnels, by means of a calibration coefficient. Applying the least squares method to the previous data leads to a calibration coefficient of 1.2066. The results of the regression are shown in 4. It has an $R^2 = 0.99996$, which indicates that this approach leads to a very good fit.

After calibrating the wind tunnel, it is necessary to calibrate the balance used to measure forces. The one used in this experiment has been previously described, so for this section it is sufficient to know that it provides force measurements on two perpendicular axes, F_x and F_y and a moment measurement around a z axis perpendicu-

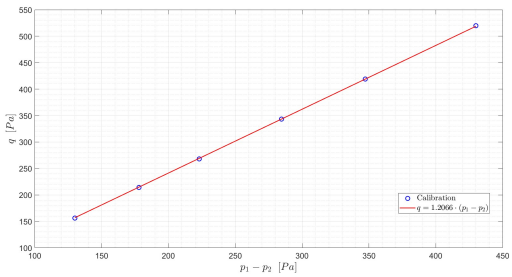

Figure 4: Wind tunnel calibration results

lar to the other two, M. An evaluation matrix E can be defined such that:

$$\vec{F} = E\vec{S}$$

Where $\vec{F}$ represents the real forces acting on the body and $\vec{S}$ represents the signals received from the balance's sensors. In order to determine this matrix, different cases are tested with known loads and a least squares solution is found for the components of E. It is good practice to select load cases that are similar to those that will be encountered in measurement. In this case, a total of four geometric configurations with two different load intensities are tested, for a total of eight cases. In all of them a weight of either 0.5 kg or 1 kg is used. Furthermore, a 20 cm offset is had in the direction perpendicular to the xy plane, to mimic the load in the center section of the model. The four configurations are as follows:

- A weight horizontally aligned with the balance's z axis, with the x axis horizontal and the y axis vertical.
- A weight horizontally aligned with the balance's z axis, with the y axis horizontal and the x axis vertical.
- A weight horizontally aligned with the balance's z axis with the x and y axis forming 45° with the vertical, such that the weight force is in the first quadrant.
- A weight with a 20 cm horizontal offset from the balance's z axis, with the x axis horizontal and the y axis vertical.

Note that the first three configurations correspond to the application of a force on each of the force axes and the simultaneous application of force on both. Finally, the fourth setup leads to the simultaneous application of a force in one axis and a moment around the z direction.

The coefficients of the evaluation matrix are determined by a least squares approach and they are computed by rows. To illustrate the procedure, consider the case of the force in the x direction. The first row of the evaluation matrix links the real F_x with the measurements of all forces and moments. Isolating this equation for all of the eight test cases, it is possible to construct the following system of equations:

$$\vec{F_x} = \bar{\bar{S}}\vec{E_1'}$$

Where $\vec{F_x}$ contains the values of force in the x direction in all cases, $\bar{\bar{S}}$ is a matrix whose rows are composed by the signals received from the balance in each test case and $\vec{E_1'}$ is the transpose of the first row of E. This system has as many rows as test cases and three columns. By solving it in the least squares sense the values for the first row of E are obtained. Repeating this procedure for F_y and M, E can be reconstructed. In this case, the results are as follows:

$$E = \begin{bmatrix} 30530 & 365.32 & -2770.8 \\ -875.57 & 30268 & 296.86 \\ 14.632 & -35.689 & 11208 \end{bmatrix}$$

Note that, regarding dimensions, E takes as an input the signals from the balance directly, and outputs loads in SI units.

Finally, the corrected values of F_x and F_z must be projected into wind axes to obtain the values of lift and drag. Using the known angles of attack, this is done as follows:

$$L = F_z\ cos(\alpha) - F_x\ sin(\alpha)$$

$$D = F_z\ sin(\alpha) + F_x\ cos(\alpha)$$

4. Experiment Results

The results of the wind tunnel experiment provide the data necessary to characterize aerodynamic behavior of the tested airfoil profile. This section presents the lift (L), drag (D), and moment (M) forces as functions of the angle of attack (α), as well as their respective dimensionless coefficients (C_L, C_D, and C_M). Additionally, the lift-drag polar plot, which is instrumental for visualizing the efficiency of the airfoil across different operating conditions, is also discussed.

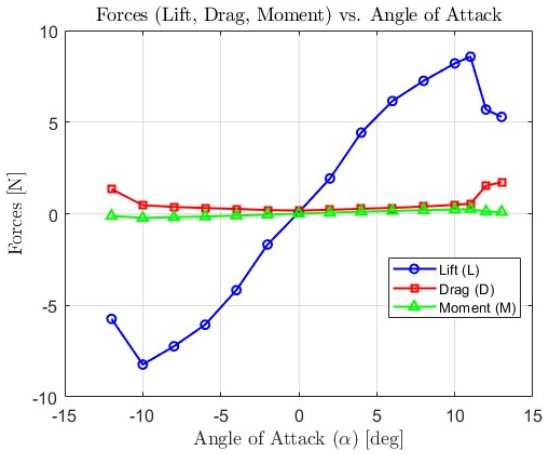

Figure 5: Forces (Lift, Drag, Moment) vs. Angle of Attack

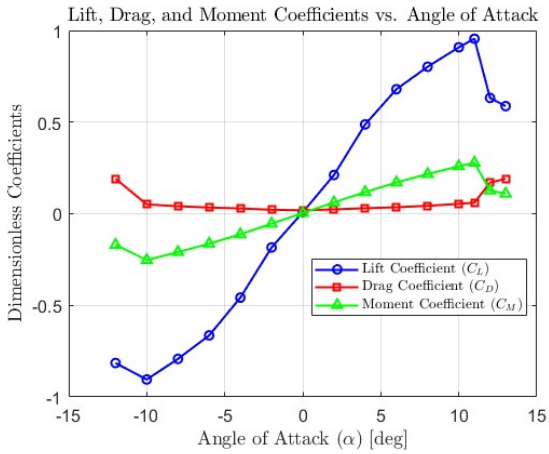

Figure 6: Lift, Drag, and Moment Coefficients vs. Angle of Attack

4.1. Forces as a Function of Angle of Attack

Figure 5 shows the variation of lift, drag, and moment forces with the angle of attack. As expected, the lift force L (represented by the blue curve) increases approximately linearly with α at low angles, achieving a peak before declining sharply near the stall angle. This behavior is typical of airfoils, where the lift increases with angle of attack until the flow separates, leading to a rapid drop in lift.

The drag force D (in red) remains relatively low and stable over the small to moderate angle range. It shows a slight increase as the angle of attack becomes positive, reflecting the increased frontal area and flow separation effects at higher angles.

The moment M (green curve) shows a more gradual change with α, staying relatively constant in the tested range. This suggests stable pitching moment characteristics for the airfoil within the angle of attack limits.

4.2. Dimensionless Coefficients (C_L, C_D, C_M) as a Function of Angle of Attack

In Figure 6, the lift (C_L), drag (C_D), and moment (C_M) coefficients are plotted against the angle of attack. The lift coefficient C_L shows a linear increase with angle of attack up to a maximum point, after which it declines sharply, indicating the stall region. This peak defines the critical angle of attack, beyond which the airfoil loses lift efficiency due to flow separation.

The drag coefficient C_D remains low at small angles of attack and gradually increases as α rises, especially near the stall region. This trend corresponds to the aerodynamic drag characteristics typical of airfoils, where drag increases significantly when the flow is less attached to the surface.

The moment coefficient C_M remains mostly stable across the angle of attack range, with a slight increase at positive angles. This stability in C_M suggests that the airfoil exhibits balanced pitching behavior across the tested angles, a desirable characteristic for many aerodynamic applications.

4.3. Lift-Drag Polar Plot

The lift-drag polar plot in Figure 7 depicts C_L as a function of C_D. This plot is useful for assessing the airfoil's aerodynamic efficiency across different angles of attack. The curve shows a distinct loop, typical of airfoil polar plots, where the highest C_L values correspond to relatively low C_D values before the stall region. The efficiency, defined as the ratio C_L/C_D, is maximized at moderate angles of attack before the stall, indicating the optimal performance range for this airfoil.

At higher C_D values (corresponding to higher angles of attack near stall), the lift efficiency declines as the airfoil experiences greater drag rel-

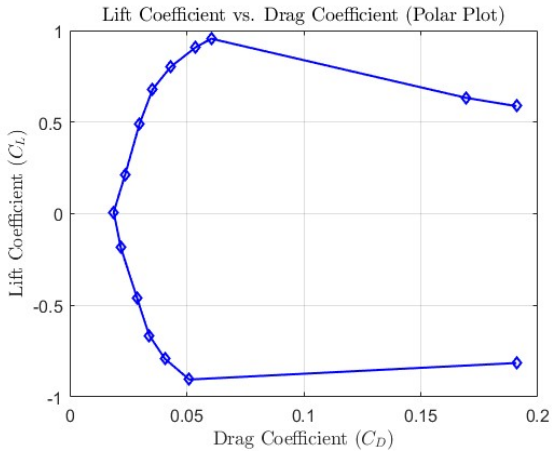

Figure 7: Lift Coefficient C_L vs. Drag Coefficient C_D (Polar Plot)

ative to lift. This decline illustrates the onset of aerodynamic inefficiency and flow separation.

4.4. Summary

Overall, the results align with typical airfoil aerodynamic characteristics, with a well-defined stall angle, efficient lift generation at moderate angles, and a stable moment coefficient across the tested range. These findings underscore the suitability of this airfoil profile for applications where stable lift and moment characteristics are crucial under varying angles of attack.

5. Conclusions

This wind tunnel experiment provided comprehensive insights into the aerodynamic behavior of a specific airfoil profile under varying angles of attack. The meticulous calibration of both the wind tunnel and the external force balance was essential to obtain accurate and reliable data. By calibrating the wind tunnel's dynamic pressure measurement with a high correlation coefficient ($R^2 = 0.99996$), a precise calibration coefficient was obtained. Furthermore, the calibration of the force balance using known weights and load configurations ensured that the measurements for lift, drag, and moment were consistent with real-world expectations. This rigorous approach minimized systematic errors and enhanced the overall accuracy of the results.

Key aerodynamic coefficients, including the lift (C_L), drag (C_D), and moment (C_M) coefficients, were computed across a range of angles of attack. The results showed typical airfoil behavior: the lift coefficient increased nearly linearly with angle of attack up to a stall point, after which flow separation caused a steep drop in lift. The drag coefficient increased steadily, especially near the stall region, while the moment coefficient remained stable, reflecting balanced pitching characteristics that are advantageous in aerodynamic applications requiring stability.

The lift-drag polar plot further highlighted the airfoil's aerodynamic efficiency, indicating an optimal lift-to-drag ratio at moderate angles of attack before the stall region. This efficient performance range demonstrates the suitability of the airfoil for applications that require both effective lift generation and low drag.

Consequently, qualitative agreement is had between the experimental and expected results. To further prove the adequacy of the chosen experimental setup, quantitative comparison with different experiments at similar Reynolds numbers would be necessary.

In conclusion, the experiment shows promise, adequately capturing the behaviour of the airfoil, at least in a quantitative sense. Quantitative validation and fine tuning of the process is left as a future line of inquiry.